Final Review

1 Module 4: Ch:7.1-7.5

1.1 Problem 1: Incentives and Wildlife Conservation

Consider two approaches to protecting elephants from poachers in African countries:

- **Plan A (Exclusion):** The government creates large national parks and forbids all local people from entering or interacting with the habitat and elephants in any way.
- **Plan B (Inclusion/Tourism):** The government creates national parks and designates 10 surrounding villages as official tourist centers, providing jobs for locals as guides and hospitality staff for park tours.

Compare the economic incentives of the local villagers under both plans. Which plan is more likely to successfully protect the elephant population in the long run? Explain using the concept of incentives.

2 Module 5:Ch:9.1-8.4

Refer Week 9 materials for problems

2.1 Problem 6: Marginal Cost and Individual Supply

A firm's marginal cost (MC) curve above the average variable cost (AVC) curve is equal to the firm's individual supply curve. This means that every time a firm receives a price from the market, it will be willing to supply the amount of output where $P = MC$.

1. What happens to the firm's individual supply curve if marginal costs increase (e.g., due to an increase in input prices)?
2. If new technology in a perfectly competitive market brings about a substantial reduction in costs of production, how will this affect the market in both the short run and the long run?

3 Module 6: Ch:9.1-9.2, 10.1-10.2, 11.1-11.3, 4

3.1 Problem 1: Market Entry and Natural Monopoly

Suppose P_0 is \$10 and P_1 is \$11. A new firm with the same LRAC curve as the incumbent tries to break into the market by selling 4,000 units of output. Refer to Figure 1 below.

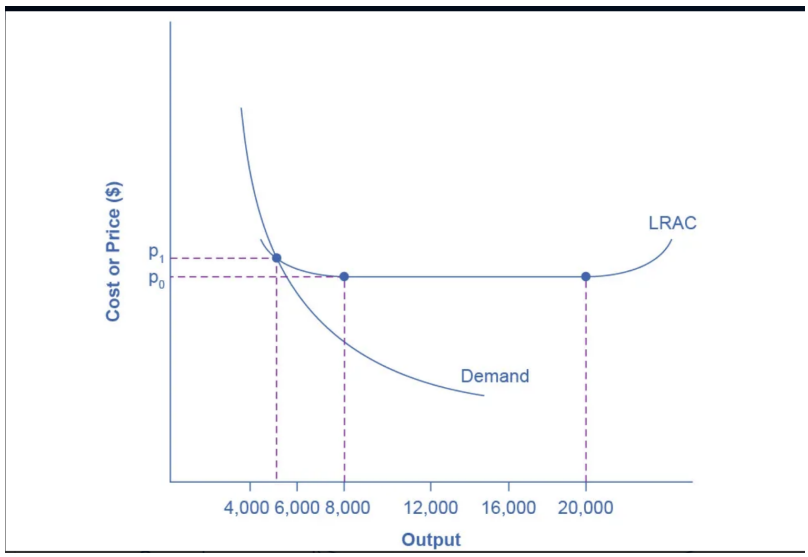

Figure 1: Long-Run Average Cost and Market Entry

1. Estimate from the graph what the new firm's average cost of producing output would be.
2. If the incumbent continues to produce 6,000 units, how much output would the two firms supply to the market?
3. Estimate what would happen to the market price as a result of the supply of both the incumbent firm and the new entrant.
4. Approximately how much profit would each firm earn?

3.2 Problem 2: Andrea's Day Spa Profit Maximization

Andrea's Day Spa began to offer a relaxing aromatherapy treatment. The firm asks you how much to charge to maximize profits. Given the demand and total cost (TC) data in Table 1, calculate the missing values and determine the profit-maximizing output.

Price ($)	Quantity	TC ($)	TR ($)	MR ($)	AC ($)	MC ($)
25.00	0	130			–	–
24.00	10	275				
23.00	20	435				
22.50	30	610				
22.00	40	800				
21.60	50	1,005				
21.20	60	1,225				

Table 1: Demand and Cost Data for Aromatherapy Treatments

1. For each level of output, calculate Total Revenue (TR), Marginal Revenue (MR), Average Cost (AC), and Marginal Cost (MC).
2. What is the profit-maximizing level of output for the treatments and how much will the firm earn in profits?

3.3 Problem 3: Classifying Barriers to Entry

Classify each of the following scenarios as a **government-enforced barrier**, a **non-government-enforced barrier**, or a situation that **does not involve a barrier to entry**.

1. A patented invention.
2. A popular but easily copied restaurant recipe.
3. An industry where economies of scale are very small compared to the size of demand in the market.
4. A well-established reputation for slashing prices in response to new entry.
5. A well-respected brand name that has been carefully built up over many years.

4 Answers

4.1 Module 5: Problem 1 - Supply Curves and Technological Change

1. **Increase in Marginal Costs:** If MC increases, the marginal cost curve shifts **upward and to the left**. Since the MC curve (above AVC) *is* the supply curve, the firm's individual supply curve effectively decreases. At any given price, the firm will now produce a **lower quantity** than before because it reaches the $P = MC$ point sooner.

2. **Technological Advancement in Perfect Competition:**

   - **Short Run:** Existing firms see their cost curves (ATC and MC) shift downward. Since $P > ATC$ for the early adopters, these firms will earn **economic profits**.
   - **Long Run:** These profits attract new firms to enter the market. As entry occurs, the **market supply curve shifts to the right**, driving the market price down.
   - **Final Result:** Entry continues until the price falls to the minimum of the new, lower ATC. In the long-run equilibrium, firms again earn zero economic profit, but consumers benefit from a **lower market price** and a **higher total quantity** produced.

4.2 Module 4: Problem 1 - Wildlife Conservation Incentives

Plan B is more likely to succeed because it aligns the economic interests of the local population with the survival of the elephants.

- **Incentives in Plan A:** Villagers often see elephants as a **liability**. Elephants may damage crops, and since the villagers gain no benefit from the park, they have no reason to report poachers. In cases of extreme poverty, villagers may even assist poachers for a share of the profit.

- **Incentives in Plan B:** Elephants become a **valuable asset**. If the villagers' livelihoods depend on tourism (guides, lodging, services), a live elephant is worth significantly more to them than a dead one.

- **Result:** In Plan B, the local community has a direct financial incentive to protect the habitat and act as "informal guards" against poaching to ensure the continued flow of tourist income.

4.3 Module 6: Problem 1

- **Total Supply:** $6,000 + 4,000 = 10,000$ units.
- **Price Effect:** The increased supply will drive the market price significantly **below** P_0.
- **Profit:** If the market price falls below the $LRAC$ at those specific quantities, both firms will likely experience economic losses, especially the smaller entrant whose AC is higher due to lower volume.

4.4 Module 6: Problem 2 - Andrea's Day Spa

To find the profit-maximizing point, we complete the table using the following formulas: $TR = P \times Q$, $MR = \frac{\Delta TR}{\Delta Q}$, $AC = \frac{TC}{Q}$, and $MC = \frac{\Delta TC}{\Delta Q}$.

P	Q	TC	TR	MR	AC	MC
25.00	0	130	0	–	–	–
24.00	10	275	240	24.00	27.50	14.50
23.00	20	435	460	22.00	21.75	16.00
22.50	30	610	675	21.50	20.33	17.50
22.00	**40**	**800**	**880**	**20.50**	**20.00**	**19.00**
21.60	50	1,005	1,080	20.00	20.10	20.50
21.20	60	1,225	1,272	19.20	20.42	22.00

Profit-Maximizing Level:

- **Rule:** Produce where $MR \geq MC$ but before MC exceeds MR.
- At $Q = 40$, $MR = \$20.50$ and $MC = \$19.00$.
- At $Q = 50$, $MC = \$20.50$ which is greater than $MR = \$20.00$.
- **Optimal Output:** 40 treatments at a price of \$22.00.
- **Total Profit:** $TR - TC = 880 - 800 =$ **\$80.00**.

4.5 Module 6: Problem 3 - Barriers to Entry

- **A patented invention:** *Government-enforced barrier*. Patents are legal protections granted by the government to provide exclusive rights to an inventor.

- **A popular but easily copied restaurant recipe:** *Not a barrier to entry*. If a recipe is easily copied, it does not prevent other firms from entering the market and competing.
- **Small economies of scale relative to demand:** *Not a barrier to entry*. This describes a competitive market where many small firms can operate efficiently without being disadvantaged by size.
- **Reputation for slashing prices (Predatory Pricing):** *Non-government-enforced barrier*. This is a strategic barrier where the incumbent uses market behavior to intimidate or price out potential entrants.
- **Well-respected brand name:** *Non-government-enforced barrier*. Brand loyalty and identity are significant hurdles for new competitors, but they are built through private investment, not government decree.